

Constrained Realization: A Cross-Level Framework for the Causal Provenance of Conscious Experience

Abstract

Psychological explanations of conscious experience span development, learning, perception, memory, motivation, affect, and neural mechanism. Prior work already establishes that similar surface outcomes can arise from different mechanisms, that intervention is central to causal explanation of consciousness, and that developmental history and plasticity shape later processing. This article integrates those established ideas into a provenance-first framework with six consciousness-relevant roles: developmental construction, present constraint, candidate-state formation, an admission/recurrent realization complex, stabilization when separable, and downstream use. Its proposed contribution is the specific localization discipline linking those roles to a negative-report causal-dissociation matrix, a measurement seam between phenomenality and report, and adjacent-depth evidence requirements. The framework also extends longitudinally: a claimed historical update must be linked to a persistent internal or environmental change and to altered later processing, while losing explanatory force if contemporaneous state variables fully capture the relevant susceptibility. No dendritic, thalamocortical, molecular, or developmental mechanism is claimed as novel or identical to phenomenality.

Keywords: consciousness; causal provenance; causal inference; developmental systems; neuroplasticity; apical amplification; reportability

1. Introduction

Conscious experience is a general-psychology problem because no single traditional subdiscipline owns its causal history. A particular episode can depend on developmental organization, learning, motivation, affect, perception, memory, bodily state, and neural dynamics operating across markedly different timescales. These literatures often measure different portions of the same causal trajectory with different constructs and methods.

Psychology and neuroscience already contain strong decompositions of parts of this trajectory. Developmental and learning theories address how organism-specific capacities are built; research on attention, motivation, affect, and memory addresses current prioritization and context. Consciousness science distinguishes neural prerequisites from candidate correlates and downstream consequences (Aru et al., 2015), while Apical Amplification Theory (AAT) and Dendritic Integration Theory (DIT) connect content-bearing activity to context-sensitive dendritic processing (Aru et al., 2020a; Marvan et al., 2021). Levels-of-analysis work already warns that apparently similar cognitive or conscious surface phenomena can arise through dissimilar mechanisms (Overgaard & Mogensen, 2011), and interventionist work applies causal-manipulation logic directly to brain-experience relations (Klein & Barron, 2020).

These advances sharply limit any claim that a new mechanism of consciousness has been discovered. The present framework makes no such claim. Its remaining question is inferential: when established developmental, cognitive, affective, and biological processes are placed on the causal trajectory of one candidate phenomenal state, does their causal depth change what an observed endpoint identifies? The framework therefore asks not which single mechanism "is" consciousness, but which causal alteration could have produced the observed result and what evidence would distinguish the alternatives.

The proposed decomposition is provenance-first. It organizes mechanisms by their causal role in producing a candidate conscious state rather than by subdiscipline, anatomical scale, or favored theory. The core roles are: (1) construction or developmental provenance; (2) present constraint; (3) candidate-state formation; (4) an admission/recurrent realization complex; (5) stabilization or persistence; and (6) downstream use through report, memory, decision, or action. These are functional explanatory roles, not claims of a fixed serial anatomy. The trajectory is also recursive: some events alter the architecture that constrains subsequent candidate formation, so provenance is not exhausted by a one-way historical chain.

The framework is motivated by an identification problem that recurs across psychological research. The general fact that one terminal outcome can be compatible with different mechanisms is established rather than novel (Overgaard & Mogensen, 2011). Icarus asks whether that general caution becomes more operational for consciousness when the alternatives are placed at six specific causal roles. Failure to report a stimulus, for example, can reflect changed motivational or contextual priority, weak or absent candidate representation, disruption of realization-related dynamics, loss of persistence, memory failure, or response failure. The same terminal behavior therefore does not, by itself, determine which role was altered or whether the phenomenal outcome was equivalent.

This article therefore develops a cross-level causal map, not an identity theory. It does not identify a dendritic calcium event, a burst, or recurrent thalamocortical activity with phenomenality. It distinguishes causal-emergence claims from identity claims. Intervention-based causal explanation is likewise not presented as novel (Klein & Barron, 2020). The narrower empirical burden is that a claim about one Icarus role should be supported by evidence that discriminates relevant neighboring alternatives rather than inferred from a terminal report or detection endpoint alone. Such evidence can constrain localization and rule out alternatives, but it is not generally sufficient without valid measures, appropriate interventions where feasible, and control of competing pathways. If the decomposition adds no inferential leverage beyond existing frameworks, it should collapse back into its established components.

2. Prior decompositions and the remaining cross-level problem

The nearest prior art spans both consciousness mechanisms and the inferential/developmental frameworks used to interpret them. Most component mechanisms, generic endpoint underidentification, interventionist causal explanation, history-dependent plasticity, and reciprocal organism-environment development are already established. The residual question is narrower: whether coordinating those precedents through one six-role consciousness-specific provenance map adds useful localization and experimental constraints.

Aru et al. (2015) distinguish prerequisites of consciousness, candidate neural correlates, and consequences of consciousness, guarding against contamination of the target process by antecedent or downstream activity. The present framework accepts that methodological separation but subdivides the causal trajectory more finely into developmental construction, current constraint, candidate formation, realization-related processing, persistence when separable, and downstream use.

At the mechanistic level, AAT and DIT already place context-sensitive apical processing and apical-basal integration near the transition between represented content and conscious processing (Aru et al., 2020a; Marvan et al., 2021). Whyte et al. (2025) further provide a burst-dependent thalamocortical model of perceptual awareness. Icarus therefore does not own candidate-to-admission biology, dendritic integration, recurrent processing, or the idea that contextual state can alter whether content reaches a stronger processing regime.

Phillips (2023) provides the broadest mechanistic collision by linking context-sensitive cooperative neuronal processing with development, learning, attention, emotion, working memory, long-term memory, and conscious experience. Storm et al. (2024) likewise establish that multiscale integration is not itself novel. These precedents remove novelty from simply arranging development, context, cellular dynamics, and recurrence on one diagram.

Overgaard and Mogensen (2011) provide a closer inferential precedent than the earlier mechanism-focused comparison acknowledged. Their consciousness-relevant multiple-realization framework explicitly distinguishes surface phenomena from underlying processing and warns that apparently similar outcomes can arise through dissimilar mechanisms. The general claim that endpoint similarity does not establish mechanistic equivalence therefore belongs to prior work; Icarus instantiates that principle in a particular candidate-conscious-state role matrix rather than claiming the principle itself.

Klein and Barron (2020) independently narrow the methodological residual. They apply interventionist causal explanation directly to consciousness and distinguish explanatory difference-makers from correlations, background or enabling conditions, precursors, and downstream effects. Intervention-based localization is therefore not an Icarus novelty. The remaining question is whether the six Icarus roles specify useful neighboring alternatives and evidentiary patterns for such interventions.

The recursive side is equally precedented. Cleeremans (2011) explicitly makes learning and plasticity central to the organization of conscious representations. Transactional developmental theory treats individual and context as reciprocally changing components of development (Sameroff, 2010), and Moore (2026) develops a current general-psychology developmental-systems synthesis in which psychological phenomena depend on present and historical context across timescales, including experience-sensitive epigenetic regulation. Reciprocal organism-environment feedback and history-dependent biological change therefore cannot serve as residual novelty.

Taken together, these precedents leave a specific integrative residual. Icarus coordinates (1) the six-role consciousness-focused causal ordering, (2) a negative-report matrix that maps one terminal endpoint onto distinct expected evidence patterns, (3) a measurement seam that keeps phenomenal occurrence distinct from representation, accessibility, memory, report, and action, (4) an adjacent-depth discrimination discipline, and (5) a longitudinal rule requiring a claimed historical update to be linked to persistent internal or environmental change and altered later consciousness-relevant processing. The framework further gives up historical provenance when contemporaneous state variables fully mediate the susceptibility of interest.

The proposed contribution is therefore organizational and inferential rather than mechanistically proprietary. Generic many-to-one causation, interventionism, developmental path dependence, plasticity, and organism-environment recursion are inputs to the framework, not discoveries of it. The paper succeeds only if their coordination through these consciousness-specific roles changes what an experiment should measure, manipulate, compare, or conclude.

Table 1. Nearest relevant precedents and the residual Icarus integration

Framework / precedent	What it already establishes	What it does not by itself establish	Relation to Icarus
Aru et al. (2015)	Prerequisite/NCC/consequence separation; downstream contamination problem	The six-role construction-to-use localization matrix	Provides the broad methodological separation
Overgaard & Mogensen (2011)	Similar surface cognitive/conscious outcomes may arise from dissimilar mechanisms; levels matter	The Icarus six-role negative-report matrix and longitudinal update rule	Establishes the generic surface-equivalence warning
Klein & Barron (2020)	Interventionist causal explanation of consciousness; background/enabling/downstream distinctions	The Icarus role taxonomy and its specific evidentiary matrix	Establishes intervention-based causal localization in principle
AAT / DIT / Whyte et al. (2025)	Context-sensitive candidate processing, dendritic integration, recurrent thalamocortical awareness models	Developmental-to-downstream provenance ordering or phenomenality identity	Provide realization-side mechanistic anchors
Phillips (2023) / Storm et al. (2024)	Development, contextual neuronal cooperation, learning/memory, conscious experience, multiscale integration	The same six-role localization matrix and longitudinal collapse rule	Broad mechanistic and multiscale precedents
Cleeremans (2011)	Learning/plasticity as central to the organization of conscious representation	The Icarus causal-role matrix and negative-report localization scheme	Owens generic plasticity-consciousness coupling
Moore (2026) / Sameroff (2010)	Historical developmental systems; reciprocal person-context change; experience-sensitive biological development	A consciousness-specific candidate/realization/report matrix	Own broad recursive developmental framing

3. Terms, scope, and causal provenance across timescales

A central measurement distinction is between physical occurrence and phenomenal occurrence. A neural state can be physically present and causally active without this framework assuming that the state is itself phenomenal. Conversely, phenomenal occurrence is not defined here by successful later report, memory, or voluntary control.

Physical presence, encoding, accessibility to a neural pathway, later reconstructability, and behavioral influence are therefore treated as distinct evidentiary properties. None of those properties by itself establishes that the relevant content was phenomenally experienced at the time under study. The distinction is methodological rather than metaphysical: it prevents a measurement channel from silently becoming a definition of the target phenomenon.

The evidence base requires an important scope restriction. The strongest mechanistic work connecting apical dendrites, layer-5 bursting, and thalamocortical dynamics to conscious processing comes from perceptual tasks and predominantly from animal studies. Dykstra et al. (2026) describe AAT and DIT as circuit-level theories for which crucial human tests remain outstanding. Accordingly, the mechanistic discussion below is strongest for perceptual consciousness. Extension of the full architecture to conscious experience generally is a theoretical proposal, not an established empirical generalization.

The provenance axis spans nested timescales. At the deepest level relevant here are inherited biological constraints and development. Across development and lifetime learning, the nervous system acquires an organism-specific architecture. At shorter timescales, current bodily and cognitive state alter the effective constraints on processing. Candidate states form and compete over still shorter intervals. Dendritic and recurrent circuit events occur over milliseconds to seconds. Report and later memory can follow after the target episode has ended. Short events can also have longer-lived consequences when they induce learning, gene regulation, synaptic modification, or changes in behavior that alter subsequent exposure. Causal influence therefore need not terminate at the downstream-use stage of one episode.

This temporal depth is central because later stages inherit the effects of earlier stages without containing a literal record of every causal precursor. A dendritic event at time t does not encode an organism's developmental history in the ordinary representational sense, yet the probability and consequences of that event depend on an architecture whose properties were built and modified by that history. Provenance therefore concerns causal dependence, not explicit online representation of the past.

For the same reason, the architecture should not be read as a strict sequence of anatomical boxes. Several processes can operate concurrently or recursively. The decomposition asks what explanatory role a mechanism performs, not whether one named structure always activates before another.

4. Construction: developmental provenance of the available system

The first causal question is not why a particular content becomes conscious. It is why this organism possesses this particular space of possible neural states, priorities, and transformations.

Brain development is neither a literal readout of a completed genetic blueprint nor a process determined independently by biology. The relevant environment is present from the beginning: prenatal conditions, sensory and social input, bodily state, spontaneous activity, and the organism's changing physiology all participate in development. This is established developmental-systems territory rather than an Icarus novelty: contemporary formulations treat psychological outcomes as products of historically extended

organism-context systems, with biological and environmental factors co-acting across timescales (Moore, 2026; Sameroff, 2010; Stiles & Jernigan, 2010).

Genetic provenance must also be separated from a fixed-genome interpretation. The genome contributes generative constraints on developmental possibilities rather than a completed adult phenotype. DNA sequence is comparatively stable, whereas transcriptional and epigenetic regulatory states can remain responsive to experience. Genetic and gene-regulatory processes therefore matter here insofar as they help construct or modify the neural architecture relevant to later processing; the framework does not treat DNA sequence as a moment-to-moment selector of conscious content.

This matters for consciousness only indirectly. Construction does not explain phenomenality. It explains why later conscious-processing mechanisms have the organization, tuning, plasticity, connectivity, and available state repertoire that they do. The mature system inherits developmental structure in sensory pathways, association cortex, memory systems, neuromodulatory systems, valuation systems, and the cellular architecture on which contextual and recurrent computations depend.

A useful abstraction is a historically conditioned space of physically available neural trajectories. Development and accumulated plasticity do not choose one present conscious state from this space; they shape which trajectories are feasible for this organism and the transition structure among them. The abstraction is intentionally qualitative. Its only commitment is that current state transitions depend on a historically constructed system rather than on the immediate stimulus alone.

Construction is therefore a provenance variable within Icarus, but historical path dependence itself is established. Two organisms receiving similar current input can differ because their architectures were built and modified differently. Even highly similar genomes do not imply identical adult organization because developmental dynamics, stochastic variation, self-generated behavior, and subsequent input can diverge. Once behavior emerges, organisms can also help alter later conditions through active and evocative environment selection (Scarr & McCartney, 1983) and broader reciprocal person-context transactions (Sameroff, 2010). Icarus uses these precedents to locate the provenance of current susceptibility rather than to claim a new developmental principle.

The falsifiable burden at this level is modest. If developmental and learning history never change the later susceptibility of candidate formation or realization mechanisms, construction would be irrelevant to the proposed causal trajectory. Existing neuroscience makes such a complete independence implausible, but the framework does not require every developmental variable to be traced to every conscious episode. The role is to specify the depth from which current susceptibility can originate.

5. Present constraint architecture

Construction specifies the available system; it does not determine the next conscious state by itself. At a particular moment the organism occupies a current configuration in which some signals and trajectories are more influential than others.

The term constraint is used here in a limited dynamical sense: a factor changes the set, probability, gain, or transition accessibility of candidate states without being identical to the candidate content itself. Current constraints can include bodily state, arousal, active goals, learned expectation, emotional valuation, environmental context, recently active representations, task demands, and memory-dependent information.

These constraints have different provenances. Developmentally instantiated constraints arise from the system's constructed organization. History-dependent constraints arise from learning and plasticity. State-dependent constraints arise from the organism's current physiological and cognitive condition. Task- or goal-dependent constraints arise from immediate commitments and control settings. The categories are analytically useful but not assumed to map one-to-one onto distinct neural modules.

Contextual information can be distributed across memory, association, interoceptive, and neuromodulatory systems without forming a mandatory serial pipeline. The provenance framework therefore does not require any fixed anatomical route from context to candidate content, and it makes no claim that a contextual source directly generates phenomenality.

AAT provides a strong cellular precedent for the more general functional idea. Marvan et al. (2021) describe apical processing as sensitive to contextual information including current needs, goals, emotions, and other sensory information. That literature therefore already establishes that context-sensitive signals can alter whether a local content representation reaches a stronger processing regime. The current framework inherits that mechanism rather than renaming it.

The provenance addition is that a present contextual signal is itself an endpoint of prior causal history. A current goal has a learning and motivational history. A learned expectation reflects earlier experience. A bodily state reflects ongoing physiological regulation. A memory signal reflects stored relations and retrieval dynamics. The effective constraint architecture is therefore implemented within the organism without being independently authored there: its organization is continually shaped by developmental and organism-environment history. Because learning and plasticity can update that architecture, and action can change later exposure, the same organism need not encounter a later event under precisely the same effective constraints or environmental conditions.

This distinction becomes experimentally useful when current constraints are manipulated while candidate-generation and realization capacity remain otherwise intact. Such interventions should preferentially change which content is favored rather than abolish the organism's general capacity for conscious processing. The prediction is not unique to this framework, but its causal placement matters for interpreting what a successful manipulation has identified.

6. Recursive provenance: experience can modify future constraint

Recursive organism-context development and learning-dependent plasticity are established ideas, including in transactional developmental theory, developmental-systems accounts, and theories that explicitly connect learning to conscious representation (Cleeremans, 2011; Moore, 2026; Sameroff, 2010). The present extension uses that established recursion for a narrower longitudinal localization problem. Let C_t denote the organism's effective constraint architecture at time t , X_t the relevant environmental and bodily conditions, and E_t an event or episode processed under those conditions. C_t and X_t jointly shape how E_t is processed. Some events produce durable learning, molecular regulation, synaptic or structural plasticity, or altered learned models, represented schematically as ΔC_t ; downstream action can alter subsequent exposure or social/environmental conditions, producing X_{t+1} . Qualitatively, $(C_t, X_t) \rightarrow E_t \rightarrow \{\Delta C_t, \text{action}\} \rightarrow (C_{t+1}, X_{t+1})$. Icarus adds no novelty to the existence of that loop. Its claim is that a historical update matters to the provenance analysis only when the update can be linked to altered processing in a later episode.

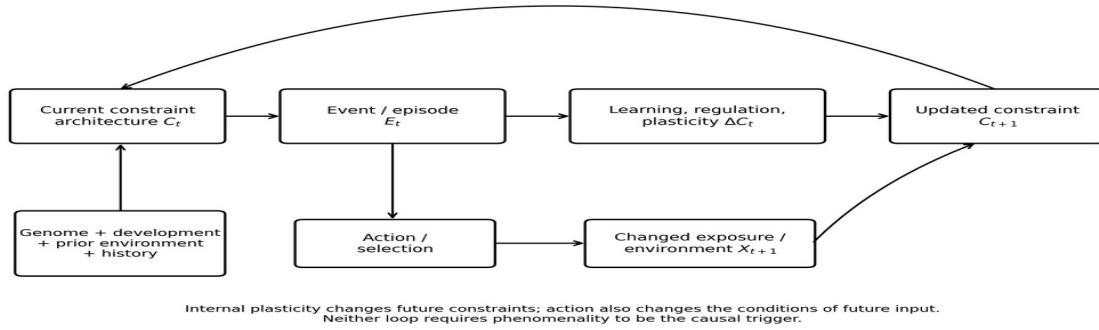


Figure 1. Recursive provenance and organism-environment coupling. Events processed under the current constraint architecture can update that architecture through learning and plasticity, while downstream action can alter subsequent exposure or environmental conditions. Both pathways can shape later processing; phenomenal occurrence is not assumed to be necessary for either update. Arrows denote possible causal classes rather than universal transitions.

This recursion requires a distinction between genomic sequence and regulatory state. The inherited DNA sequence is comparatively stable, whereas methylation, demethylation, chromatin state, transcription, and related regulatory processes can remain experience-sensitive in mature neural systems. In adult rats, contextual fear learning transiently increased hippocampal DNA-methyltransferase expression, altered methylation of memory-related genes, and was disrupted by inhibition of DNA methyltransferases (Miller & Sweatt, 2007). At a longer developmental timescale, maternal-care differences in rats were associated with persistent methylation and expression differences at a hippocampal glucocorticoid-receptor promoter, with cross-fostering altering the developmental outcome (Weaver et al., 2004). Human postmortem hippocampal data provide convergent but observational evidence: childhood-abuse history has been associated with altered NR3C1 promoter methylation and glucocorticoid-receptor expression (McGowan et al., 2009). These findings do not make DNA a moment-to-moment selector of consciousness. They show that molecular regulation is one route by which history can become part of the machinery shaping future susceptibility.

The hippocampus is relevant here as a history-sensitive contextual system, not as a mandatory gateway through which all sensory input must pass. Hippocampal and connected circuits support relational and episodic context, novelty and mismatch, pattern separation and completion, and retrieval of learned relations. Molecular change in these systems can therefore alter how later information is contextualized or weighted. Acute stress, for example, can produce dentate-gyrus-specific changes in DNA methylation and immediate-early gene regulation, with manipulations of methyl-donor availability altering later molecular and behavioral responses (Saunderson et al., 2016). Environmental enrichment provides a nonpathological complement: genetically homogeneous mice raised in the same enriched environment developed persistent individualized behavior accompanied by differences in hippocampal neurogenesis and DNA methylation (Zocher et al., 2020). The causal point is not that one molecular mark determines a later experience, but that the architecture carrying prior history remains physically editable.

Psychedelic perturbations provide a useful rapid-plasticity example because acute changes in neural organization can be separated from longer-lived changes that outlast drug presence. In mice, a single psilocybin exposure increased formation and size of dendritic spines on layer-5 pyramidal neurons within about a day, with a subset of structural changes persisting for weeks (Shao et al., 2021). A single exposure

to the serotonergic psychedelic DOI produced longer-lived synaptic and epigenomic changes together with altered fear-extinction behavior (de la Fuente Revenga et al., 2021). In humans, dense longitudinal neuroimaging has shown large acute changes in functional-network organization after psilocybin together with more persistent changes in some connectivity measures over subsequent weeks (Siegel et al., 2024). These findings do not establish that the psychedelic subjective state causes the persistent changes, nor do they identify a mechanism of phenomenality. They illustrate a bounded intervention in which acute circuit perturbation can be followed by longer-timescale changes relevant to future constraint architecture.

Shared inherited sequence likewise does not fix a shared developmental trajectory. Monozygotic twins can diverge epigenetically across life (Fraga et al., 2005), and prospective twin work has associated differential childhood/adolescent stress histories with later within-pair personality differences while remaining observational and limited in sample size (Torgersen & Janson, 2002). Genetically identical mice in a shared enriched environment can also develop persistent individualized behavior accompanied by differences in hippocampal neurogenesis and DNA methylation (Zocher et al., 2020). These examples support historical divergence under similar starting conditions; they do not provide a consciousness-specific mechanism.

Learned world models provide a psychological route for the same recursive logic. Long-term priors can alter perceptual interpretation through measurable top-down cortical interactions (Hardstone et al., 2021). Goals, expectations, religious or metaphysical commitments, beliefs about personal identity, and other higher-order models can therefore matter causally when they are physically represented and alter attention, expectation, valuation, emotion, action, or environmental exposure. The framework makes no claim that a nonphysical metaphysical entity acts on neurons. Its claim is that a belief about the world can become an acquired constraint on how subsequent information is interpreted and what actions are selected. Those actions can then change the evidence, situations, and social responses encountered later, creating a second feedback path through the organism's environment. Such models can therefore become self-reinforcing or can be revised by discrepant experience.

Recursive provenance is compatible with both continuity and bounded change. Behavioral dispositions, preferences, goals, and other relatively stable tendencies can reflect accumulated genomic, developmental, stochastic, learning, and state-dependent influences while remaining partly modifiable. Plasticity is nevertheless constrained by existing anatomy, connectivity, receptor distributions, age, prior learning, physiological state, disease or injury, environmental access, and the circuits recruited by a new event. Not predetermined is therefore not equivalent to infinitely malleable.

For Icarus, the longitudinal implication is therefore deliberately narrower than the generic claim that history changes organisms. A prior event earns a place in the causal provenance of a later candidate only when evidence links that event to a persistent internal or environmental update and links that update to changed later susceptibility, weighting, candidate formation, realization-related processing, or relevant behavior. If contemporaneous state variables fully capture that later susceptibility, the historical path adds genealogy rather than an additional explanatory level. The recursive extension thus closes the loop around the original six-role decomposition without becoming a second theory of development.

7. Candidate-state formation

Constraint is not content. A candidate state is a presently instantiated neural state that can participate in further competition, amplification, recurrence, or downstream processing without being assumed phenomenal merely because it exists.

The distinction between representation and conscious experience is substantially preceded. AAT explicitly allows perceptual content to be processed without awareness and proposes that sufficient apical amplification can alter whether that content receives conscious status (Marvan et al., 2021). The broader literature on unconscious processing likewise demonstrates that behavioral or neural sensitivity can occur without ordinary reports of awareness. The present framework does not claim to discover this distinction.

Candidate formation is retained as a separate causal role because it permits a clean intervention question: is the relevant content present in a measurable neural form before the proposed realization mechanism is altered? If not, a later failure cannot be attributed specifically to admission into conscious processing. If a candidate is measurable while the realization complex is selectively perturbed, stronger causal localization becomes possible.

A candidate need not be a single localized representation. It can be distributed, recurrent, partial, competitive, or internally generated. The concept is positional rather than anatomical. It marks a state for which there is evidence of content-bearing neural organization before one has established phenomenal occurrence.

The framework therefore preserves a conservative relation:

Representation or candidate presence does not entail phenomenal experience.

This also prevents retrospective relabeling. When a participant later reports a stimulus, every antecedent process should not automatically be called conscious merely because it contributed to a conscious outcome. Candidate formation is precisely the stage at which content can be physically real and causally relevant while the phenomenal status of that content remains open.

8. The admission/recurrent realization complex

The most mechanistically specific part of the framework concerns the transition from candidate processing into the neural regime associated with conscious perception. The relevant biology is not novel to this proposal.

Layer-5 pyramidal neurons integrate inputs arriving at different cellular compartments. Larkum et al. (1999) demonstrated a nonlinear mechanism by which coincident inputs to different cortical layers can interact through dendritic calcium electrogenesis and alter somatic output. This work established a cellular basis for coupling feed-forward and apical inputs rather than treating the neuron as a simple point integrator.

Takahashi et al. (2016) later showed that active cortical dendrites causally modulate perceptual detection in mice. Takahashi et al. (2020) further showed that active dendritic currents in layer-5 pyramidal-tract neurons can gate descending cortical outputs during tactile perception, including outputs to higher-order thalamic and subcortical targets. These experiments support a causal role for apical dendritic state in perceptual performance, but they do not by themselves establish that a dendritic event is phenomenality.

AAT and DIT provide the closest consciousness-level interpretations. AAT proposes that apical amplification can grant selected perceptual content conscious status (Marvan et al., 2021). DIT proposes that appropriate integration of apical and basal compartments is central to conscious processing (Aru et al., 2020a). General anesthesia can disrupt coupling between apical and somatic compartments while leaving some feed-forward responsiveness comparatively intact, providing an additional dissociation between basic sensory responsiveness and integrated pyramidal-cell processing (Suzuki & Larkum, 2020).

Whyte et al. (2025) extend the candidate mechanism into an explicit thalamocortical model. In their model, burst-dependent interactions between cortex and higher-order thalamus support dynamical regimes associated with perceptual awareness and reproduce several causal and behavioral observations. This provides a mechanistic bridge between local cellular nonlinearities and recurrent circuit dynamics, but it remains a computational model rather than proof that the modeled regime is identical to conscious experience.

The present framework groups this family of events under the deliberately noncommittal term realization complex. The complex can include apical-basal interaction, regenerative dendritic activity, layer-5 burst output, higher-order thalamic recruitment, and recurrent cortical-thalamic dynamics. The phrase “realization” denotes a transition from a candidate possibility into an actually instantiated neural regime. It does not, by itself, explain why that regime is phenomenal.

Three relations remain unresolved. First, a regenerative dendritic event could be part of the earliest neural transition associated with conscious admission. Second, it could merely initiate a larger recurrent process whose organization is the relevant substrate. Third, the distinction between local admission and subsequent recurrence may be artificial if the mechanism is fundamentally a coupled dynamical process. Current evidence does not justify an identity claim such as “dendritic calcium spike equals consciousness.”

Accordingly, the strongest statement allowed by the evidence is causal rather than metaphysical: these cellular and circuit mechanisms are serious candidates for processes that alter whether represented perceptual content enters and persists in a neural regime associated with conscious perception.

9. Stabilization and persistence

The realization complex raises a second question: should initial admission and subsequent stabilization be treated as distinct causal roles?

Conceptually, admission asks what changes when a candidate first enters the relevant processing regime. Stabilization asks what permits that regime to persist, dominate competing activity, or remain available over time. Whyte et al. (2025) make recurrent thalamocortical dynamics central to sustained perceptual states, which makes a stabilization role plausible. However, the available evidence does not establish that there is a temporally or mechanistically discrete admission event followed by a separately manipulable maintenance event.

The framework therefore treats admission-versus-stabilization as an empirical decomposition with an explicit collapse condition. If interventions can preserve an early candidate-to-realization transition while selectively reducing persistence, dominance duration, or resistance to replacement, the distinction gains support. If every manipulation of recurrence also abolishes all credible markers of initial admission, the two roles may not be separately identifiable and should collapse into one realization process.

This uncertainty matters because a theoretical diagram can easily create artificial stages simply by drawing separate boxes. The present framework does not count a stage as scientifically useful merely because it can be named. Stabilization remains in the framework only because it generates a discriminating intervention question.

10. Externally anchored and internally generated content

A causal model of conscious realization should not be restricted to externally driven perception. Conscious experience can occur when external sensory anchoring is weak, altered, or absent, as in dreaming and some anesthetic states. Therefore candidate formation and realization must be separable from the external source of the candidate content.

Aru et al. (2020b) propose “apical drive” as a possible cellular mechanism for dreaming, extending apical-function ideas from amplification of externally driven sensory input toward internally generated experience. The proposal is not established as a mechanism of dreaming, but it demonstrates that the same cellular architecture can be considered in regimes where internally generated activity supplies more of the effective drive.

This supports a useful decomposition:

source of content is not the same causal question as realization of content.

An externally anchored candidate may be driven strongly by current sensory input. An internally generated candidate may depend more heavily on memory, association, spontaneous dynamics, imagery, or bodily signals. Both could, in principle, converge on overlapping realization machinery. Conversely, a manipulation that reduces external connectedness does not by itself demonstrate loss of phenomenal experience.

This distinction also limits the manuscript's empirical claims. Evidence from perceptual detection tasks can support a mechanism of perceptual admission without automatically establishing that all forms of conscious thought, imagery, dream experience, or emotion use the same implementation. Generalization beyond perception remains unestablished and should be tested rather than assumed.

11. Downstream use and the measurement seam

Consciousness experiments frequently rely on what happens after the target episode. A participant reports a stimulus, makes a discrimination, forms a memory, or changes behavior. These outcomes are indispensable evidence, but they introduce additional causal steps.

The present framework groups such effects under downstream use. This category includes report, deliberate decision, memory formation or retrieval, and overt action when those processes occur after or depend on the target state. Downstream use is not assumed to be wholly post-phenomenal in every case; recurrent interactions between reporting systems and perceptual systems can occur during a task. The category is instead a warning against treating the final measurement channel as identical to the target phenomenon.

Aru et al. (2015) make the related point that consequences of consciousness can contaminate attempts to isolate the neural correlates of consciousness. The provenance-first formulation extends that warning in both directions. A negative report can result from an alteration before candidate formation, within the

realization complex, during stabilization, or in the reporting pathway itself. A later reconstruction can also contain information that was not necessarily used online in the same form during the target episode.

The evidentiary asymmetry is therefore important. A reliable positive report can support the occurrence of experience under ordinary conditions, but failure to report does not uniquely identify absence of experience. It may indicate absence, but that conclusion requires bridging assumptions about candidate formation, realization, persistence, memory, and response capacity.

This is a measurement seam rather than a claim that reports are untrustworthy. The point is causal: information being present, encoded, decodable, reportable later, or reconstructable in a later episode does not automatically establish how that information participated in the causal trajectory at the time under study.

12. Causal dissociations, observational underidentification, and psychological inference

The causal-dissociation matrix in Table 2 is the operational core of the provenance-first decomposition. The general warning that similar surface outcomes can conceal different mechanisms is established (Overgaard & Mogensen, 2011). Table 2 is the Icarus-specific instantiation: it maps one negative-report endpoint onto distinct expected evidence patterns at current constraint, candidate formation, realization-related processing, persistence, and downstream use.

Let $R(X)$ denote successful report of target content X . An observation of not- $R(X)$ does not identify one hidden cause. At minimum, several causal families remain possible: current constraints may have favored competing content; a sufficiently strong candidate representation of X may never have formed; a candidate may have formed but failed to enter the realization complex; a realization-related transition may have occurred but failed to persist; or downstream memory/report mechanisms may have failed.

These possibilities are not merely different stories about the same mechanism. Each row of the matrix implies a different evidentiary pattern and a different next measurement. A candidate-level alteration predicts weak or absent content-bearing representation. A realization-level alteration predicts measurable candidate content with altered admission-related dynamics. A stabilization-level alteration predicts evidence of an initial transition with reduced persistence, if that dissociation exists. A downstream-use alteration predicts comparatively preserved upstream dynamics with impaired behavioral expression or later access.

The observation is therefore underidentified with respect to the Icarus roles, but generic endpoint underidentification is not claimed as novel. Behavioral equivalence does not establish mechanistic equivalence, and mechanistic nonequivalence prevents automatic phenomenal equivalence. The matrix contributes only insofar as its role-specific alternatives change the next measurement, intervention, or interpretation. It does not prove that experience survived any particular negative-report trial.

The same logic applies to positive behavior. Two participants can produce the same correct response through different mixtures of sensory evidence, contextual priors, candidate dynamics, recurrent stability, and downstream decision processes. The framework therefore treats endpoint agreement as insufficient for causal localization and favors designs that measure or manipulate more than one depth before attributing an effect to conscious processing itself.

Table 2. Causal-dissociation matrix: one negative-report endpoint, multiple causal depths

Altered causal depth	Candidate-content evidence	Realization-related evidence	Persistence	Report	Interpretive consequence
Current constraint	Changed or deprioritized	Competent	Competent	Changed/absent	Different content favored; no localization to realization
Candidate formation	Absent or altered	Competent in principle	Competent in principle	Changed/absent	Content may never reach the relevant representation
Realization complex	Present	Impaired	Often impaired	Changed/absent	Candidate can exist without entering the target processing regime
Stabilization (if separable)	Initially present	Early transition may occur	Impaired	Weak/absent	Tests whether admission and persistence can dissociate
Downstream use	Present	Present	Present	Impaired	Endpoint does not identify whether phenomenality was absent

13. Intervention classes and discriminating consequences

Interventionist causal explanation is already established in consciousness methodology (Klein & Barron, 2020). The matrix earns additional scientific value only if its specific role distinctions change which variables should be manipulated or measured and which alternatives remain viable. The following intervention classes therefore operationalize the Icarus decomposition rather than introduce interventionism itself.

Construction- and update-level interventions alter the learned or developmental organization of the system over extended timescales. Their predicted signature is a change in the repertoire, weighting, or susceptibility of later states rather than a momentary global loss of consciousness. This class includes developmental perturbations, durable learning-induced changes, and bounded interventions that leave measurable molecular, synaptic, structural, or representational changes after the initiating event has ended. Doron et al. (2020), for example, show that perirhinal input to neocortical layer 1 controls learning and is associated with changes in dendritic excitability. The recursive extension adds a longitudinal requirement: when an event is claimed to have altered future constraint architecture, evidence should connect the event to a persistent change at the proposed update level and then to altered processing at a later episode.

Constraint-level interventions alter current context, expectation, goal, bodily state, or related signals while leaving candidate-generation and realization capacity otherwise available. Their primary predicted effect is a change in which content is favored. AAT and other contextual accounts already provide strong

precedent for this prediction; the provenance framework's contribution is to localize the interpretation of the manipulation rather than claim the effect as new.

Candidate-level interventions alter the content-bearing representation itself. If realization machinery remains competent, the expected result is changed or absent content rather than a generalized failure of conscious capacity. Candidate measurements are also necessary controls for realization-level claims: one cannot conclude that admission failed if the relevant candidate was never present.

Realization-level interventions seek the stronger dissociation: candidate representation remains measurable while cellular or circuit events associated with conscious perception are selectively impaired. Takahashi et al. (2016, 2020), Suzuki and Larkum (2020), and the AAT/DIT literature provide important precedents for this logic, although the behavioral and anesthetic endpoints do not by themselves resolve phenomenality.

Stabilization-level interventions remain the least established class. A useful experiment would preserve an early realization-related transition while selectively changing duration, dominance, or resistance to replacement. Failure to identify such a dissociation would support collapsing stabilization into the realization complex.

Downstream-use interventions alter memory, reporting, or action while upstream candidate and realization-related dynamics remain comparatively intact. Such experiments are difficult to interpret because positive evidence of preserved phenomenality often depends on downstream channels. The framework therefore predicts underidentification rather than assuming preserved experience by definition.

A discriminating empirical program should combine interventions and measurements across at least two adjacent causal depths. Candidate representation should be measured when realization is perturbed; early realization markers should be measured when recurrence is perturbed; and upstream dynamics should be measured when report is perturbed. Such adjacent-depth evidence can constrain localization and rule out some neighboring alternatives, but it is not generally sufficient for causal identification by itself. Strong attribution also depends on measurement validity, selective interventions where feasible, and control of alternative pathways. A single terminal behavior is insufficient.

14. Constrained realization as a cross-level state-space interpretation

The phrase constrained realization summarizes the architecture but should not be treated as an explanation by itself.

At any time, a biological system occupies a physical state within a historically constructed space of accessible transitions. Development, plasticity, and current context change which trajectories are feasible or favored. A candidate state is an actual neural configuration within that space. The realization complex refers to a further physical transition into a processing regime associated with conscious perception. In this sense, constraint answers why some trajectories are more available than others, while realization refers to what physically occurs rather than what was merely possible.

The state-space language is intentionally qualitative. It does not specify an energy metric, neural code, or solved transition equation. Cellular events such as dendritic electrogenesis matter here because they provide mechanistically grounded instances of state transition; their physical reality does not imply phenomenal identity.

The strongest defensible compression is therefore not an empirical identity but a research hypothesis: conscious episodes may be understood as highly structured physical realizations occurring within organism-specific constraint architectures. The relevant biological realization might involve the apical-burst-thalamocortical family of mechanisms discussed above, but the exact phenomenal boundary remains unresolved.

Accordingly, “consciousness = constrained realization” should be read as a theoretical summary of the decomposition, not as a solved equation. If future work identifies phenomenality with a different mechanism, the provenance architecture could remain useful while the proposed realization implementation changes. If the decomposition itself fails to improve causal attribution, even the summary should be abandoned.

15. Loss, collapse, and subsumption conditions

The framework is designed to lose components when the evidence does not support them.

First, admission and stabilization should collapse into one realization process if they cannot be independently manipulated or measured. Separate labels are not sufficient evidence for separate mechanisms.

Second, candidate and realization should collapse if no candidate content can be identified independently of the dynamics proposed to realize it. The representation-realization distinction is scientifically useful only when a measurable antecedent state can be separated from the altered processing regime.

Third, the provenance claim loses scientific force if developmental and historical variables add no explanatory leverage beyond current-state descriptions. If all relevant susceptibility can be fully captured by current variables without reference to how the architecture was constructed, provenance becomes historical narrative rather than a causal variable for the target analysis.

The recursive extension adds related loss conditions. If purported history-dependent updates never improve prediction or interpretation of later processing once the later current state is measured, then the feedback loop is descriptive history rather than an additional causal level. Molecular or structural change is not relevant merely because it persists; it must alter later susceptibility, weighting, transition, or behavior implicated in the analysis. Likewise, action-mediated environmental feedback contributes nothing distinctive if it adds no predictive or interpretive leverage beyond established ecological, learning, transactional, or developmental-systems models.

Fourth, the framework is fully subsumed if an established theory already contains the same six-role construction-to-constraint-to-candidate-to-realization-to-stabilization-to-use decomposition together with an equivalent negative-report evidence matrix, phenomenality/report seam, adjacent-depth discrimination discipline, and longitudinal update-to-later-processing rule. The present literature contains close partial precedents across those elements, including Aru et al. (2015), Overgaard and Mogensen (2011), Cleeremans (2011), AAT/DIT, Klein and Barron (2020), Phillips (2023), Storm et al. (2024), Whyte et al. (2025), and Moore (2026). The targeted comparison does not establish a single source or tightly connected framework containing the complete integration.

Fifth, the framework fails if causal depth does not change experimental interpretation. A diagram that merely sorts known mechanisms without altering what observations support is not a substantive theoretical contribution.

Finally, no result within this framework should be described as identifying consciousness itself unless it distinguishes a causal relation to phenomenal occurrence from an identity relation. Evidence that mechanism Q is necessary for report, detection, or recurrent processing supports $Q \rightarrow \text{outcome}$ under specified conditions. It does not by itself establish Q is identical to phenomenality.

16. Scope and limitations

Several limitations sharply bound the claims of the present framework.

The strongest mechanistic evidence concerns perceptual detection and perceptual awareness. Generalizing from these tasks to all conscious thought, emotion, dreaming, pain, or self-experience remains unresolved. The internally generated-content section provides a reason to test generalization, not evidence that it has succeeded.

Most direct causal work on apical dendrites comes from animal models. Human circuit-level tests of AAT and DIT remain an active methodological challenge (Dykstra et al., 2026). Behavioral detection in animals cannot by itself adjudicate whether a manipulated state was phenomenally experienced.

The recursive-provenance evidence is similarly heterogeneous. Causal demonstrations of experience-dependent methylation and dendritic plasticity are strongest in animal models; human hippocampal epigenetic findings are largely observational, and human psychedelic studies do not presently provide a before-and-after neural epigenomic chain linking subjective experience to later conscious processing. Peripheral molecular changes would not by themselves establish neural modification. The framework therefore uses these findings to support the physical editability of constraint architecture, not to claim a molecular marker of consciousness.

The provenance stages are functional roles, not guaranteed anatomical modules. The same neural structure can contribute to multiple roles, and several roles can overlap in time. The architecture is therefore not a claim that consciousness proceeds through a fixed serial pathway.

The state-space description is qualitative rather than a measured high-dimensional manifold supplied by the present paper. Quantitative implementation would require an explicit state definition, measurement model, and transition dynamics.

Plasticity also does not imply arbitrary self-modification. Existing anatomy, developmental history, connectivity, receptor distributions, age, physiological state, pathology, environmental access, and the circuits recruited by a particular event constrain both the direction and magnitude of change. The framework therefore predicts path dependence and bounded malleability, not that personality or preference can be deliberately rewritten at will.

The framework does not solve the explanatory gap between neural dynamics and phenomenality. It deliberately leaves open whether phenomenality begins with a local nonlinear event, a recurrent circuit regime, some broader organization, or another mechanism not considered here. Its contribution is to improve the causal bookkeeping around that unresolved boundary.

Finally, the novelty claim is synthetic and compositional. The component mechanisms and the broad methodological principles of multiple realization, interventionism, developmental path dependence, plasticity, and reciprocal organism-environment change are established. The manuscript should therefore be evaluated only on whether its specific six-role consciousness localization architecture, evidence

matrix, measurement seam, and longitudinal collapse rule improve experimental design and interpretation.

17. Conclusion

A particular conscious state is the endpoint of a causal trajectory operating across nested timescales and across the traditional boundaries of psychological subdisciplines. Development constructs an organism-specific system through continuing interaction among inherited biology, environment, bodily state, stochastic variation, and prior activity. Learning, motivation, affect, memory, and ongoing physiology modify the effective constraints within that system. Current constraints influence which candidate states become competitive. Cellular and recurrent circuit mechanisms alter whether candidate content enters and persists within a neural regime associated with conscious perception. Report, memory, decision, and action introduce additional downstream processes. Crucially, some events then feed back into the system through learning, molecular regulation, synaptic or structural plasticity, learned models, and altered environmental trajectories, changing both the internal constraints and external conditions under which later candidates arise.

None of these components is proposed as a newly discovered mechanism, and neither generic endpoint underidentification nor recursive development is claimed as new. The residual proposal is their specific integration into a six-role consciousness-focused localization architecture: developmental construction, present constraint, candidate formation, realization-related processing, stabilization when separable, and downstream use.

The general principle that similar endpoints can conceal different mechanisms is already established (Overgaard & Mogensen, 2011). Icarus makes that principle concrete for a candidate conscious episode by coupling the six roles to a negative-report evidence matrix and a phenomenality/report measurement seam. The matrix is useful only when neighboring evidence constrains causal localization more than terminal behavior alone.

Constrained realization is the compression of this view. Constraint describes how a historically constructed organism narrows and weights possible trajectories; realization describes an actual transition into a neural regime associated with conscious processing. Longitudinally, an earlier event belongs to the later provenance analysis only when it leaves a persistent internal or environmental update that changes subsequent consciousness-relevant processing; if contemporaneous state fully mediates that susceptibility, the historical path loses independent explanatory force. Where phenomenality begins within the relevant neural regime remains unresolved.

The framework succeeds only if causal depth changes what experiments mean. If it does not improve attribution across psychological and neural levels, the mosaic should collapse back into its already established components.

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